Installing MySQL Server

To Begin with the Lab

Summary of the Lab

This lab covered installing and managing MySQL Server on Ubuntu, verifying and removing the installation, and running MySQL inside a Docker container. A sample database and employee table were created with test data. A PHP-Apache container was then built to connect to the MySQL container using the MySQLi extension and display employee records through a web application, demonstrating successful containerized database and application integration.

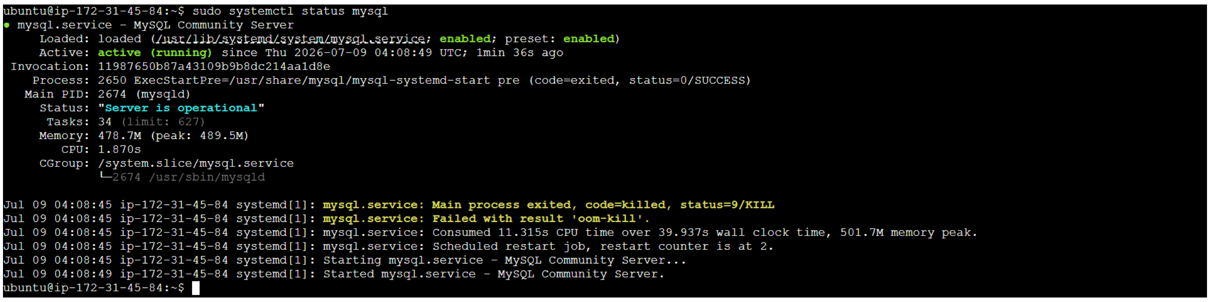
- Install MySQL Server

```
sudo apt update

sudo apt install mysql-server -y
```

- Verify MySQL

```
sudo systemctl status mysql
```



- Start MySQL

```
sudo systemctl start mysql
```

- Open the MySQL Shell

```
sudo mysql
```

```
ubuntu@ip-172-31-45-84:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.4.10-0ubuntu0.26.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.03 sec)

mysql>
```

- Now delete the mysql and its dependencies
- For the first stop the mysql services

```
sudo systemctl stop mysql
```

- then Remove MySQL packages

```
sudo apt purge mysql-server mysql-client mysql-common mysql-server-core-* mysql-client-core-* -y
```

- Remove automatically installed dependencies

```
sudo apt autoremove --purge -y
```

```
ubuntu@ip-172-31-45-84:~$ sudo apt autoremove --purge -y
REMOVING:
  libcgi-fast-perl*  libfcgi-bin*   libfcgi0t64*             libhtml-template-perl*  libprotobuf-lite32t64*   mysql-client-core*
  libcgi-pm-perl*    libfcgi-perl*  libgoogle-perftools4t64*  libmecab2*              libtcmalloc-minimal4t64*  mysql-server-core*

Summary:
  Upgrading: 0, Installing: 0, Removing: 12, Not Upgrading: 22
  Freed space: 181 MB

(Reading database ... 166630 files and directories currently installed.)
Removing libcgi-fast-perl (1:2.17-1) ...
Removing libhtml-template-perl (2.97-2build1) ...
Removing libcgi-pm-perl (4.71-1build1) ...
Removing libfcgi-bin (2.4.5-0.1build2) ...
Removing libfcgi-perl (0.82+ds-3build3) ...
Removing libfcgi0t64:amd64 (2.4.5-0.1build2) ...
Removing mysql-server-core (8.4.10-0ubuntu0.26.04.1) ...
Removing mysql-client-core (8.4.10-0ubuntu0.26.04.1) ...
Removing libgoogle-perftools4t64:amd64 (2.18.1-1) ...
Removing libmecab2:amd64 (0.996-15.1build4) ...
Removing libprotobuf-lite32t64:amd64 (3.21.12-15ubuntu1) ...
Removing libtcmalloc-minimal4t64:amd64 (2.18.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
(Reading database ... 166365 files and directories currently installed.)
Purging configuration files for libmecab2:amd64 (0.996-15.1build4) ...
ubuntu@ip-172-31-45-84:~$
```

- Running MySQL in a Docker Container
- A custom container name, Port mapping, Root password

```
sudo docker run -d --name mysql-instance -p 3306:3306 -e MYSQL_ROOT_PASSWORD=1234 mysql
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker run -d --name mysql-instance -p 3306:3306 -e MYSQL_ROOT_PASSWORD=1234 mysql
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
6b21eb7a1e3e: Pull complete
d7fddb948e8a: Pull complete
8ae77f7c30a4: Pull complete
80b86c090661: Pull complete
5caad7b89f9f: Pull complete
047567c900ab: Pull complete
8a1152ace190: Pull complete
3737f34f1214: Pull complete
143f320bd2c8: Pull complete
33485abd4a2e: Pull complete
35c8d9e39e0c: Download complete
86b43b4257d4: Download complete
Digest: sha256:ad88e1c86cbf12ef52d0a0360cd4b774d22956c5ee9c565645fb019d7aacb6c3
Status: Downloaded newer image for mysql:latest
6adf8f9db818168c883eb8b0ff9fd02ea1d6958027316d72e16ea5d1fb52acca
ubuntu@ip-172-31-45-84:~$
```

- Verify the Container

```
sudo docker ps
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                                    NAMES
6adf8f9db818   mysql     "docker-entrypoint.s…"   3 minutes ago   Up 3 minutes   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp   mysql-instance
ubuntu@ip-172-31-45-84:~$
```

- Install the MySQL Client
- Update the package index.

```
sudo apt update
```

- Install the MySQL client.

```
sudo apt install mysql-client -y
```

- Connect to the MySQL Container

```
mysql -h 127.0.0.1 -u root -p
```

- Enter the password:

```
ubuntu@ip-172-31-45-84:~$ mysql -h 127.0.0.1 -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.1 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

- Connect a PHP Container to a MySQL Container

Note: This lab assumes you are using the official PHP-Apache image and a MySQL container. No custom Docker image is built in this exercise.

- Go to mysql
- Create a Sample Database
- Connect to MySQL.

```
mysql -h 127.0.0.1 -u root -p
```

- Enter your password.
- Create a Database

```
CREATE DATABASE company_db;
```

- Use it.

```
USE company_db;
```

```
ubuntu@ip-172-31-45-84:~/app1$ cd ..
ubuntu@ip-172-31-45-84:~$ mysql -h 127.0.0.1 -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.1 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE company_db;
Query OK, 1 row affected (0.03 sec)

mysql> USE company_db;
Database changed
mysql>
```

- Create a Table

```
CREATE TABLE employees (

    employee_id INT PRIMARY KEY AUTO_INCREMENT,

    employee_name VARCHAR(100),

    department VARCHAR(50),

    salary DECIMAL(10,2)

);
```

```
mysql> CREATE TABLE employees (
    ->
    ->     employee_id INT PRIMARY KEY AUTO_INCREMENT,
    ->
    ->     employee_name VARCHAR(100),
    ->
    ->     department VARCHAR(50),
    ->
    ->     salary DECIMAL(10,2)
    ->
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
```

- Insert Sample Data

```
INSERT INTO employees
```

(employee_name, department, salary)

VALUES

('Aman','Cloud',65000),

('Rahul','DevOps',72000),

('Priya','AI/ML',85000);

```
mysql> INSERT INTO employees
    -> (employee_name, department, salary)
    -> VALUES
    ->
    -> ('Aman','Cloud',65000),
    ->
    -> ('Rahul','DevOps',72000),
    ->
    -> ('Priya','AI/ML',85000);
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM employees;
+-------------+---------------+------------+----------+
| employee_id | employee_name | department | salary   |
+-------------+---------------+------------+----------+
|           1 | Aman          | Cloud      | 65000.00 |
|           2 | Rahul         | DevOps     | 72000.00 |
|           3 | Priya         | AI/ML      | 85000.00 |
+-------------+---------------+------------+----------+
3 rows in set (0.00 sec)

mysql>
```

- Create a new folder.
- Go the folder and create a Dockerfile and a index.php file.

```
ubuntu@ip-172-31-45-84:~$ mkdir app1
ubuntu@ip-172-31-45-84:~$ cd app1
```

- Create index.php file
- So here to get the host run the command

sudo docker inspect mysql-instance

```
        },
        "Networks": {
            "bridge": {
                "IPAMConfig": null,
                "Links": null,
                "Aliases": null,
                "DriverOpts": null,
                "GwPriority": 0,
                "NetworkID": "60b010e5e5c35989ed0f767f033e9e7d5e75ce19830813cafb289e031ca584df",
                "EndpointID": "77e1979a5685c231ad60b470e8f7adff344d73ec8e1ede1a5db5c38e94ef293c",
                "Gateway": "172.17.0.1",
                "IPAddress": "172.17.0.2",
                "MacAddress": "06:d0:7d:14:01:f5",
                "IPPrefixLen": 16,
                "IPv6Gateway": "",
                "GlobalIPv6Address": "",
                "GlobalIPv6PrefixLen": 0,
                "DNSNames": null
            }
        }
```

<?php

// Database Configuration

$host = "172.17.0.2";   // MySQL Container IP

```
$username = "root";
$password = "1234";
$database = "company_db";

// Create MySQL Connection
$conn = new mysqli($host, $username, $password, $database);

// Check Connection
if ($conn->connect_error) {
    die("<h2>Connection Failed:</h2> " . $conn->connect_error);
}

// SQL Query
$sql = "SELECT employee_id, employee_name, department, salary FROM employees";
$result = $conn->query($sql);
?>

<!DOCTYPE html>
<html>
<head>
    <title>Employee Details</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background: #f4f4f4;
            margin: 40px;
        }

        h2 {
```

```
    text-align: center;
    color: #333;
}

table {
    width: 70%;
    margin: auto;
    border-collapse: collapse;
    background: white;
}

th {
    background: #007BFF;
    color: white;
    padding: 10px;
}

td {
    padding: 10px;
    text-align: center;
}

table, th, td {
    border: 1px solid #ccc;
}

tr:nth-child(even) {
    background: #f2f2f2;
}
```

```
        .success {
            text-align: center;
            color: green;
            margin-bottom: 20px;
        }
    </style>
</head>
<body>

<h2>Employee Details</h2>

<div class="success">
    Connected successfully to MySQL Database!
</div>

<?php
if ($result->num_rows > 0) {

    echo "<table>";
    echo "<tr>
            <th>Employee ID</th>
            <th>Employee Name</th>
            <th>Department</th>
            <th>Salary</th>
          </tr>";

    while($row = $result->fetch_assoc()) {
        echo "<tr>";
```

```
        echo "<td>" . $row["employee_id"] . "</td>";
        echo "<td>" . $row["employee_name"] . "</td>";
        echo "<td>" . $row["department"] . "</td>";
        echo "<td>₹" . number_format($row["salary"], 2) . "</td>";
        echo "</tr>";
    }

    echo "</table>";

} else {
    echo "<p style='text-align:center;color:red;'>No employee records found.</p>";
}

$conn->close();
?>

</body>
</html>
```

- Now create the Dockerfile

```
FROM php:8.2-apache
COPY . /var/www/html
RUN docker-php-ext-install mysqli
```

- Now build the image from the current directory

```
sudo docker build -t php-app:v1 .
```

```
find . -name .libs -a -type d|xargs rm -rf
rm -f libphp.la      modules/* libs/*
rm -f ext/opcache/jit/zend_jit_x86.c
rm -f ext/opcache/jit/zend_jit_arm64.c
rm -f ext/opcache/minilua
 ---> Removed intermediate container 7b2640f37118
 ---> d3f1519fbcc6
Successfully built d3f1519fbcc6
Successfully tagged php-app:v1
ubuntu@ip-172-31-45-84:~/app1$
```

- Verify the image

```
sudo docker images
```

```
ubuntu@ip-172-31-45-84:~/app1$ sudo docker images

IMAGE            ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:latest     ad88e1c86cbf       1.28GB          288MB
php-app:v1       d3f1519fbcc6        708MB          176MB
ubuntu@ip-172-31-45-84:~/app1$
```

- Run the PHP container
- Since Apache listens on port **80** inside the container, map it to a port on your EC2 instance, for example **8080**:

```
sudo docker run -d --name php-app-instance -p 80:80 php-app:v1
```

- Verify the container is running

```
sudo docker ps
```

```
ubuntu@ip-172-31-45-84:~/app1$ sudo docker run -d --name php-app-instance -p 80:80 php-app:v1
aaef7e0d6cd6aa8d010ad9f1ce54842618ff7700d6a250e5cd638410e1e3ca0f
ubuntu@ip-172-31-45-84:~/app1$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS                                   NAMES
aaef7e0d6cd6   php-app:v1     "docker-php-entrypoi…"   16 seconds ago       Up 15 seconds       0.0.0.0:80->80/tcp, [::]:80->80/tcp     php-app-instance
68b327f972f6   mysql:latest   "docker-entrypoint.s…"   About a minute ago   Up About a minute   3306/tcp, 33060/tcp                     mysql-instance
ubuntu@ip-172-31-45-84:~/app1$
```

- Access the Application
- Open:
- http://<EC2-Public-IP>/index.php
- The application should display the employee data stored in the MySQL container.

Employee Details

Connected successfully to MySQL Database!

Employee ID	Employee Name	Department	Salary
1	Aman	Cloud	₹65,000.00
2	Rahul	DevOps	₹72,000.00
3	Priya	AI/ML	₹85,000.00